EGRE 245 Engineering Programming
Fall 2025
Mark Schwitzerlett

Lab Exercise #8: Thin Metal Plate Steady State Temperature

1 Overview

In this lab, you will write a C program to simulate the steady-state temperature distribution on a thin metal plate with different edge temperatures. The temperature at each interior point is determined by averaging the temperatures of its four neighboring cells. The process repeats until the plate reaches thermal equilibrium (steady state).

The grid with boundary conditions is given below.

100	100	100	100	100
90	T(1,1)	T(1,2)	T(1,3)	95
90	T(2,1)	T(2,2)	T(2,3)	95
90	T(3,1)	T(3,2)	T(3,3)	95
80	80	80	80	80

At each iteration, the interior temperatures update according to:
$T(i,j) = (T(i-1,j) + T(i+1,j) + T(i,j-1) + T(i,j+1))/4$

Your program should

- Use two separate two-dimensional arrays. One to hold the current values of the grid temperatures and a second to compute the new temperature at each successive time step.
- Use a loop to update the temperature array until every cell converges using the following method.
  1. Set a boolean variable.
  2. Copy the updated temperatures to the current temperature array.
  3. Loop over all the interior cells (but not the edge cells).
  4. Repeat this process until the temperature in every cell is within the allowable tolerance (0.01). For example, once the change in the temperature in cell $T(1,1)$ is less than or equal to 0.01 that is the final temperature for that cell.
- Track the number of iterations required.
- Print out the number of iterations and final grid to the screen.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #8 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #8 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Lab #8 -- Mark Schwitzerlett and Leonard Euler
October 21, 2025

Steady-state reached after 18 iterations.

Final temperature distribution:
 90.00 100.00 100.00 100.00  95.00
 90.00  93.92  95.21  95.71  95.00
 90.00  90.48  91.23  92.62  95.00
 90.00  86.78  86.64  88.56  95.00
 90.00  80.00  80.00  80.00  95.00


...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single ".c" file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named Lab#8_FirstName_LastName.c

Due date: The lab is due at the end of your lab period, either Tuesday, October 21th or Thursday, October 23th.

Be sure to save a copy of your .c file.